

Torque Arm Dimensions

The torque arm is used to transfer floating-caliper reaction force to the load cell. The calibrated effective lever arm length is 100.0 mm. The arm is inclined at 40 degrees relative to the horizontal to accommodate the brake caliper geometry. Torque is calculated as $T = F \times L \times \cos(40 \text{ deg})$.

Parameter	Value	Tolerance/Note
Lever arm length	100.0 mm	+/- 0.1 mm
Torque arm inclination	40 degrees	+/- 1 degree
Load cell orientation	Vertical	Measure vertical reaction force
Torque equation	$T = F \times 0.1 \times \cos(40 \text{ deg})$	$T = F \times 0.0766 \text{ m}$